

Flexible Formation and Obstacle Avoidance for multi-UAV System via Nutcracker Optimization and Trajectory Control

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Abstract: This paper proposes a flexible formation and obstacle avoidance scheme for a multi-drone system. The compound task of the two assignments is transformed into a trajectory tracking issue through path planning based on the nutcracker optimization algorithm (NOA) under a weighted communication topology. The dynamics of UAV is established to build a fully actuated error system. By designing the control law, the goal of trajectory tracking is achieved. Stability analysis and proof are provided together with simulation verification of a six-drone team to verify the effectiveness of the proposed methods.

Key Words: Fully actuated system, Formation control, Obstacle avoidance, NOA, Multi-UAV

1 Introduction

Multiple unmanned aerial vehicles (Multi-UAVs) system refers to multiple UAVs completing tasks. In recent decades, multi-UAVs systems have been highly concerned in wide applications [1]. Formation control and obstacle avoidance are important for the multi-UAVs systems. Recently, researchers have conducted a lot of research on formation control, such as the leader follower model [2], the virtual structure [3], the behavior-based model [4], etc. The leader-follower model is a popular approach owing to its simplicity and stability [5]. Moreover, in order to guarantee the implementation of formation, several other issues should be considered, for example, the obstacle and collision avoidance.

Numerous algorithms already exist for path planning and obstacle avoidance. As for drones, the algorithms are mainly divided into three categories: the optimization based methods, the potential field based methods, and the machine learning based obstacle avoidance methods [6]. The optimization-based algorithms, due to their flexibility and diversity, have garnered widespread attention from researchers. Within this category, they are further classified into mathematical optimization algorithms [7], graph theory based algorithms [8], and heuristic algorithms [9]. The mathematical optimization approach to solving models is intuitive; however, when constraints become complex, the computational load increases, making it unsuitable for real-time applications. The graph-theory based one may encounter issues such as local optimal solutions, prolonged search times, or the generation of redundant paths. The real-time performance of heuristic algorithms can be enhanced by superior optimization algorithms. There are already some intelligent algorithms used in drone path planning to achieve obstacle avoidance, such as the cuckoo algorithm [10], the whale optimization algorithm [11], the ant lion algorithm [12], and the pigeon swarm algorithm [13], etc. Among the meta-heuristic, the nutcracker optimization

algorithm [14] exhibits excellent comprehensive performance in multiple test set functions and engineering problems. Inspired by [14], in this work, we explore the application of the nutcracker method for the path planning and obstacle avoidance.

The fully actuated systems theory provides new methods for control systems [15]. Such methods linearizes high-order fully actuated (HOFA) systems to simplify the controller design [16]. As a result, with properly chosen nonlinear state feedback control laws, a linear steady-state closed-loop system with expected characteristic structure can be obtained, ensuring stable convergence of corresponding engineering problems.

In this paper, a flexible path planning and formation control strategy is proposed combining the fully actuated system control approach, the NOA algorithm for a multi-UAV formation operating within a weighted communication network. This approach provides several advantages, including a straightforward control law design, rapid convergence rates, and minimal errors.

The main contributions of this paper are

(1) A feasible and flexible strategy for obstacle avoidance of multi-agent formation is proposed based on the nutcracker optimization algorithm (NOA) over weighted communication topology.

(2) For each UAV, the other UAVs are simplified into 7 points each in three-dimensional space for obstacle avoidance. This induces the calculation amount relating to the avoidance among the UAVs moving along the road.

(3) The formation and obstacle avoidance compound tasks are transformed into a trajectory tracking issue through the fully actuated system (FAS) control law. And fully actuated system guarantees fast convergence rate, simple design, and small errors.

The rest of this paper is organized as follows. Section 2 demonstrates the UAV dynamics, the fully actuated system control approach and the nutcracker optimization algorithm. Section 3 illustrated the proposed strategy of path planning and obstacle avoidance, the control law design and the stability analysis. In section 4, we set up synthetic scenarios to validate the algorithm and discuss the results. Finally, con-

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clusions are summarized in section 5.

Notations: $\|\cdot\|$ defines the norm of vectors; $\mathbf{0}$ represents zero matrix; I is identity matrix.

2 Problem Statement and Preliminaries

2.1 UAV Dynamics Model

Quadcopter is an under-actuated system as it has six degrees of freedom but only four actual inputs. The six degrees of freedom include linear motion in three directions and rotational one around the x, y, z axes. The configuration of drones is as in Fig. 1. The earth coordinate system and the body coordinate system are represented by $Ox_e y_e z_e$ and $Ox_b y_b z_b$, respectively. Four rotors driven by a motor each, are mounted at two orthogonal directions and rotate in a direction as the circular arrow shows. For the sake of simplicity [17], we make the following definitions to divide the control input of the UAV dynamics into U_i ($i \in [1, 4]$) as follows

$$\begin{cases} U_1 = \tau_1 + \tau_2 + \tau_3 + \tau_4 \\ U_2 = -\tau_2 + \tau_4 \\ U_3 = -\tau_1 + \tau_3 \\ U_4 = -\tau_1 + \tau_2 - \tau_3 + \tau_4 \end{cases}$$

where τ_i ($i \in [1, 4]$) represents the combined thrust or force provided by the j th motor. Define $a = [\phi, \theta, \psi]^T \in \mathbb{R}^3$,

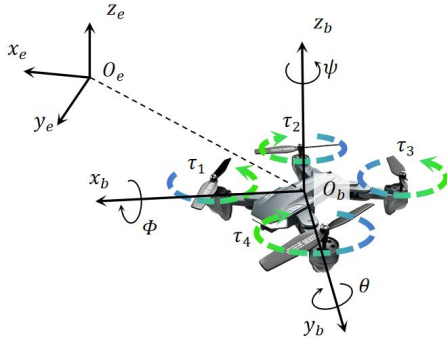


Fig. 1: Moment structure of a quadcopter drone

where ϕ, θ and ψ denote the angles of roll, pitch, and yaw with respect to the inertia frame $Ox_e y_e z_e$. Define $\dot{a} = [\dot{\phi}, \dot{\theta}, \dot{\psi}]^T \in \mathbb{R}^3$, where $\dot{\phi}, \dot{\theta}$ and $\dot{\psi}$ denote the angular velocity of roll, pitch, and yaw. $p = [x, y, z]^T \in \mathbb{R}^3$ is the position vector. Based on the conversion relationship between the ground coordinate system $Ox_e y_e z_e$ and the body coordinate system $Ox_b y_b z_b$, a rotation matrix R is introduced as

$$R = \begin{bmatrix} C_\psi C_\theta & C_\psi S_\theta S_\phi - S_\psi C_\phi & C_\psi S_\theta C_\phi + S_\psi S_\phi \\ S_\psi C_\theta & S_\psi S_\theta S_\phi + C_\psi C_\phi & S_\psi S_\theta C_\phi - C_\psi S_\phi \\ -S_\theta & C_\theta S_\phi & C_\theta C_\phi \end{bmatrix}$$

In the body coordinate system, the total lift of the UAV is expressed as:

$$U_b = [U_x \ U_y \ U_z]^T = [0 \ 0 \ U_1]^T$$

The expression of the lift force applied to the earth coordinate system is:

$$U_e = R U_b = \begin{bmatrix} C_\phi S_\theta C_\psi + S_\phi S_\psi \\ C_\phi S_\theta S_\psi - S_\phi C_\psi \\ C_\phi C_\theta \end{bmatrix} U_1$$

Based on Newton-Euler's law [18], the UAV dynamics equation can be obtained

$$\begin{cases} \ddot{x} = (\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) \frac{U_1}{m} - \frac{d_x}{m} \dot{x} \\ \ddot{y} = (\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) \frac{U_1}{m} - \frac{d_y}{m} \dot{y} \\ \ddot{z} = (\cos \phi \cos \theta) \frac{U_1}{m} - \frac{d_z}{m} \dot{z} - g \\ \ddot{\phi} = \frac{J_y - J_z}{J_x} \dot{\theta} \dot{\psi} - \frac{d_\phi}{J_x} \dot{\phi} + \frac{U_2}{J_x} \\ \ddot{\theta} = \frac{J_z - J_x}{J_y} \dot{\phi} \dot{\psi} - \frac{d_\theta}{J_y} \dot{\theta} + \frac{U_3}{J_y} \\ \ddot{\psi} = \frac{J_x - J_y}{J_z} \dot{\phi} \dot{\theta} - \frac{d_\psi}{J_z} \dot{\psi} + \frac{U_4}{J_z} \end{cases} \quad (1)$$

where g is the gravity acceleration; J_x, J_y and J_z are the moments of inertia around x, y and z axes; d_x, d_y and d_z are the air resistance coefficients; d_ϕ, d_θ and d_ψ are aerodynamic drag coefficients; m is the mass of the UAV.

Remark 1. The roll, pitch and yaw attitudes are normally limited to $(-\frac{\pi}{2}, \frac{\pi}{2})$ since the UAV has physical constraint in reality considering for safety issues.

Additionally, we make some assumptions about the UAVs:

- (1) There is no communication delay between the UAVs.
- (2) Each drone can detect its own pose, velocity and acceleration.
- (3) Drones can transmit their information mentioned above to the corresponding recipients (neighbors or followers) through communication.
- (4) There is no disturbance or fault in each drone, or that they can be reasonably ignored.

2.2 Fully Actuated System Theory Control Approach

The fully actuated system (FAS) approach [19–21] directly designs parameters for the whole high-order systems, which can ensure both the states and their derivatives converge to zero according to Hurwitz criterion. Moreover, the control law designed through FAS approach is flexible and intuitive [22].

Define a second-order nonlinear system

$$E \ddot{x} = f(x, \dot{x}, t) + B(x, \dot{x}, t) u, \quad (2)$$

where $E \in \mathbb{R}^{n \times n}$ is a matrix that can be singular; $x \in \mathbb{R}^{n \times 1}$ is the state; $f(x, \dot{x}, t) \in \mathbb{R}^{n \times 1}$ is a continuous vector function; $B(x, \dot{x}, t) \in \mathbb{R}^{n \times n}$ is the matrix of coefficients; $u \in \mathbb{R}^{n \times 1}$ is the control law to be designed. the system (2) is fully actuated with the control distribution matrix B satisfying

$$\det B(x, \dot{x}, t) \neq 0, \forall x, \dot{x} \in \mathbb{R}^{n \times 1}, t \geq 0,$$

The following linear and constant system is:

$$E \ddot{x} + A_1 \dot{x} + A_0 x = 0,$$

through selecting matrices $A_1, A_2 \in \mathbb{R}^{n \times n}$, which makes the control law as

$$u = -B^{-1}(A_1 \dot{x} + A_0 x + f),$$

Especially, let $E = I_n$. Then the system (2) is transformed into

$$\ddot{x} = -A_1\dot{x} - A_0x.$$

Define $z_1 = x$, $z_2 = \dot{x}$ and then we have

$$\dot{z} = \begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{0} & I \\ -A_0 & -A_1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = Az. \quad (3)$$

Select A_1 and A_0 properly so that the matrix A in (3) can be Hurwitz. Then the system (3) is stable consequently.

2.3 The Nutcracker Optimization Algorithm (NOA)

Inspired by Clark's nutcrackers, nutcracker optimization algorithm (NOA) is a novel nature-inspired meta-heuristic [14]. NOA is proposed to mimic the various behaviors of the special birds to present a robust meta-heuristic algorithm with different local and global search operators. It shows highly-performance on standard test functions, test suites of CEC-2014, CEC-2017, and CEC-2020 and several real-world engineering design problems. Therefore, this paper applies it into the strategy of obstacle avoidance in formation to optimize for better balance between obstacle avoidance and formation assignments.

3 Main Results

3.1 Path Planning and Optimized Obstacle Avoidance Strategy

The formation team includes one drone being the leader and others as the followers. The leader is set for leading the drones to move towards the destination, while followers need to follow it and maintain a designated relative position with neighbors to achieve the formation shape. There exist two kinds of communication: unidirectional and bidirectional. Unidirectional is applied when the leader transforms its acceleration, velocity, position to its followers, while bidirectional is used when neighbors send their own information aforementioned to each other. After calculating the weight of information from unidirectional and bidirectional, individuals execute its tracking strategies. For each of the drones, the path planning is as follows

$$p_{id}(t+1) = \alpha \cdot p_d + \beta \cdot \sum_{j \in N_i} \omega_{i,j} \cdot (p_j(t) + \Delta p_{i,j}),$$

where α and β are the parameters that mean whether the current drone i is the leader or the follower, namely

$$\begin{cases} \alpha = 1, \beta = 0, & i \text{ is the leader} \\ \alpha = 0, \beta = 1, & i \text{ is the follower} \end{cases}$$

and $p_{id}(t+1)$ is the desired position of the i th drone at the moment of $t+1$; p_d is the destination of the formation flight; N_i is the neighbors of the i th drone; $\omega_{i,j}$ is the weight of information from the j th one to the i th; $\Delta p_{i,j}$ is the relative position maintained from the j th drone to the i th; $p_j(t)$ is the position of the j th at the moment of t .

This paper considers each obstacle in the environment as a set of points within a three-dimensional spatial range. To achieve obstacle avoidance, the drone needs to maintain a certain safe distance from points in the obstacle spatial

range. In addition, for a certain drone, other drones are considered as dynamic obstacles. Considering the radius of the drone, for example, set to 0.2m, each drone will be simplified as an object occupying a set of 7 spatial points as shown in Fig. 2. Then other drones can detect whether they will collide with this drone or they maintain a safe distance from it. The obstacle avoidance is implemented through the NOA

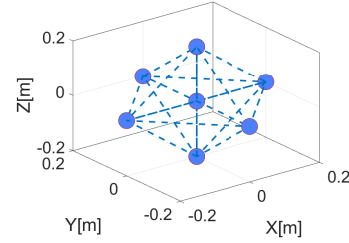


Fig. 2: Seven points that represent the space occupied by the drone

optimized algorithm, whose generation process of initializing the population is described by the following equation:

$$X_m(t) = rand \cdot (ub - lb) + lb + p_i(t), m = 1, 2, \dots, M$$

where X_m represents the position of the m th nutcracker in current iteration t ; $rand$ generates a random number between 0 and 1; ub and lb represent the upper and lower bounds, respectively; $p_i(t)$ is the current position of the i th drone; M is the number of nutcrackers in the population. X_m will be optimized for the suitable position plan of the i th drone through the NOA. The complete process of NOA is not repeated in this work due to the length of this paper. Interested readers can refer to [14] to get a deeper insight. We next illustrate the application of NOA.

This paper uses the function $NOA(\cdot)$ to represent the whole optimize progress aforementioned and the expression is

$$NOA(x) = \left\{ \arg \min_x \varrho(x) \right\}$$

where $\varrho(x)$ is the fitness function. We define the safe distance for the obstacle avoidance as

$$\eta_i(t) = \varepsilon \cdot (1 + \mu \cdot \ln(1 + \|v_i(t)\|)) \quad (4)$$

where ε is a fixed positive value representing the minimum safety distance that needs to be maintained when stationary; μ is an adjustable parameter; $v_i(t)$ is the linear velocity of the i th drone. For the i th drone, the fitness function $\varrho(\cdot)$ is

$$\varrho_i(p_{id}^*(t+1)) = \|p_{id}^*(t) - p_{id}(t+1)\| + \Xi(p_{id}^*(t+1))$$

with

$$\Xi(p_{id}^*(t+1)) = \begin{cases} \infty, & p_{id}^*(t+1) \in \gamma_{oi} \\ 0, & p_{id}^*(t+1) \notin \gamma_{oi} \end{cases}$$

The equation of γ_{oi} is

$$\gamma_{oi} = \{(x, y, z) \mid \|[x, y, z] - [x_o, y_o, z_o]\| < \eta_i(t)\},$$

where $[x_o, y_o, z_o]$ represents any obstacle point. Therefore, the obstacle avoidance and path planing strategy is expressed

as

$$p_i^*(t+1) = \begin{cases} p_{id}(t+1), & p_{id}(t+1) \notin \gamma_{oi} \\ NOA(p_{id}(t+1)), & p_{id}(t+1) \in \gamma_{oi} \end{cases}$$

Then the i th drone is ready for tracking $p_i^*(t+1)$.

3.2 Control Law Design

For the sake of convenience, the dynamics (1) is recalled as

$$\begin{cases} \ddot{x} = (\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) \frac{U_1}{m} - \frac{d_x}{m} \dot{x} \\ \ddot{y} = (\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) \frac{U_1}{m} - \frac{d_y}{m} \dot{y} \\ \ddot{z} = (\cos \phi \cos \theta) \frac{U_1}{m} - \frac{d_z}{m} \dot{z} - g \\ \ddot{\phi} = \frac{J_y - J_z}{J_x} \dot{\theta} \dot{\psi} - \frac{d_\phi}{J_x} \dot{\phi} + \frac{U_2}{J_x} \\ \ddot{\theta} = \frac{J_z - J_x}{J_y} \dot{\phi} \dot{\psi} - \frac{d_\theta}{J_y} \dot{\theta} + \frac{U_3}{J_y} \\ \ddot{\psi} = \frac{J_x - J_y}{J_z} \dot{\phi} \dot{\theta} - \frac{d_\psi}{J_z} \dot{\psi} + \frac{U_4}{J_z} \end{cases} \quad (5)$$

(1) Position control

The error vector in the position layer is

$$e_p = \begin{bmatrix} x - x_r \\ y - y_r \\ z - z_r \end{bmatrix}, \quad (6)$$

with the first and second derivatives as

$$\dot{e}_p = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix}, \quad \ddot{e}_p = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix}. \quad (7)$$

Then the fully actuated state z_p is defined as

$$z_p = \begin{bmatrix} e_p \\ \dot{e}_p \end{bmatrix},$$

whose derivative is

$$\dot{z}_p = \begin{bmatrix} \dot{e}_p \\ \ddot{e}_p \end{bmatrix} = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix} \begin{bmatrix} e_p \\ \dot{e}_p \end{bmatrix} + \begin{bmatrix} 0 \\ \ddot{e}_p \end{bmatrix}. \quad (8)$$

The error vector in position layer is

$$\ddot{e}_p = -A_1 \dot{e}_p - A_0 e_p. \quad (9)$$

And (8) is transformed into

$$\dot{z}_p = \begin{bmatrix} \dot{e}_p \\ \ddot{e}_p \end{bmatrix} = \begin{bmatrix} 0 & I \\ -A_1 & -A_0 \end{bmatrix} \begin{bmatrix} e_p \\ \dot{e}_p \end{bmatrix} = A z_p \quad (10)$$

The two matrices A_1 and A_0 can be chosen to make A be Hurwitz. Then the system (10) is stable based on Routh-Hurwitz stability criterion. With (5), (7) and (9), the following equation can be obtained

$$\begin{aligned} U_p &= \begin{bmatrix} (\cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi) U_1 \\ (\cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi) U_1 \\ (\cos \phi \cos \theta) U_1 \end{bmatrix} \\ &= -A_1 \dot{e}_p - A_0 e_p + \begin{bmatrix} \frac{d_x}{m} \dot{x} \\ \frac{d_y}{m} \dot{y} \\ \frac{d_z}{m} \dot{z} + g \end{bmatrix} \end{aligned} \quad (11)$$

Considering Remark. 1 and properties of sine and cosine functions, we can calculate that

$$\begin{cases} \|U_1\| = \text{norm}(U_p) \\ \text{sgn}(U_1) = \text{sgn}(U_p(3)) \end{cases} \quad (12)$$

Subsequently, we can solve the three reference angles ϕ_r, θ_r, ψ_r with the U_1 solved from (11).

(2) Attitude control

Similarly with (6), we define the error vector in the attitude layer as

$$e_a = \begin{bmatrix} \phi - \phi_r \\ \theta - \theta_r \\ \psi - \psi_r \end{bmatrix}.$$

With Remark. 2, the first and second derivatives are simplified as

$$\dot{e}_a = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}, \quad \ddot{e}_a = \begin{bmatrix} \ddot{\phi} \\ \ddot{\theta} \\ \ddot{\psi} \end{bmatrix}. \quad (13)$$

Remark 2. Generally, the attitude angles of UAV are small and change very slowly. Therefore, their reference derivatives $[\dot{\phi}_r, \dot{\theta}_r, \dot{\psi}_r]$ and $[\ddot{\phi}_r, \ddot{\theta}_r, \ddot{\psi}_r]$ with respect to time are considered to be approximately 0.

Then we define the fully actuated state z_a as

$$z_a = \begin{bmatrix} e_a \\ \dot{e}_a \end{bmatrix}$$

The derivative is

$$\dot{z}_a = \begin{bmatrix} \dot{e}_a \\ \ddot{e}_a \end{bmatrix} = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix} \begin{bmatrix} e_a \\ \dot{e}_a \end{bmatrix} + \begin{bmatrix} 0 \\ \ddot{e}_a \end{bmatrix}. \quad (14)$$

With

$$\ddot{e}_a = -\bar{A}_1 \dot{e}_a - \bar{A}_0 e_a, \quad (15)$$

(14) is transformed into

$$\dot{z}_a = \begin{bmatrix} \dot{e}_a \\ \ddot{e}_a \end{bmatrix} = \begin{bmatrix} 0 & I \\ -\bar{A}_1 & -\bar{A}_0 \end{bmatrix} \begin{bmatrix} e_a \\ \dot{e}_a \end{bmatrix} = \bar{A} z_a \quad (16)$$

The coefficient matrix $\bar{A}$ can be chosen to be Hurwitz, which guarantees (16) is stable. With (5), (13) and (15), we can obtain that

$$\begin{bmatrix} \frac{U_2}{J_x} \\ \frac{U_3}{J_y} \\ \frac{U_4}{J_z} \end{bmatrix} = -\bar{A}_1 \dot{e}_a - \bar{A}_0 e_a - \begin{bmatrix} \frac{J_y - J_z}{J_x} \dot{\theta} \dot{\psi} - \frac{d_\phi}{J_x} \dot{\phi} \\ \frac{J_z - J_x}{J_y} \dot{\phi} \dot{\psi} - \frac{d_\theta}{J_y} \dot{\theta} \\ \frac{J_x - J_y}{J_z} \dot{\phi} \dot{\theta} - \frac{d_\psi}{J_z} \dot{\psi} \end{bmatrix}. \quad (17)$$

With the solutions (12) and (17), we can control the drones to move towards the desired path planned.

3.3 Stability Analysis

The systems obtained in position and attitude layers are

$$\dot{z}_p = \begin{bmatrix} 0 & I \\ -A_0 & -A_1 \end{bmatrix} \begin{bmatrix} e_p \\ \dot{e}_p \end{bmatrix}, \quad (18)$$

$$\dot{z}_a = \begin{bmatrix} 0 & I \\ -\bar{A}_0 & -\bar{A}_1 \end{bmatrix} \begin{bmatrix} e_a \\ \dot{e}_a \end{bmatrix}. \quad (19)$$

Combining (18) and (19), the derivative of z is

$$\dot{z} = \begin{bmatrix} \dot{z}_p \\ \dot{z}_a \end{bmatrix} = A^* z, \quad (20)$$

where

$$A^* = \begin{bmatrix} \mathbf{0} & I & \mathbf{0} \\ -A_0 & -A_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -\bar{A}_0 & -\bar{A}_1 \end{bmatrix}.$$

The solution of linear time-invariant system (20) is given by

$$z(t) = e^{A^*(t-t_0)} z(t_0).$$

Since matrix A^* is Hurwitz, it satisfies

$$\|e^{A^*(t-t_0)}\| \leq ke^{-\lambda(t-t_0)}. \quad (21)$$

Then we can conclude that

$$\|z(t)\| \leq ke^{-\lambda(t-t_0)} \|z(t_0)\|,$$

which means

$$\lim_{t \rightarrow \infty} z(t) = 0.$$

Therefore, the error system (20) is stable.

4 Simulation Verification

To justify the performance of the proposed drones obstacle avoidance strategy and formation controller design, several simulations based on a multi-UAV system were conducted. Consider a multi-UAV system that contains six UAVs whose dynamics are expressed as (1) with the radius being 0.2m. Their communication weight topology diagram is illustrated by Fig. 3a. The formation shape is designed as in Fig. 3b, where one unit length is set to be 0.5m. In the environment where two different obstacles exist, drones need to move in a fixed formation to the goal point $[10, 10, 2]^T$. The parameters involved are set as in Table 1. The matrices

Table 1: Parameters and values

Parameter	Value
ε	0.2m
m	1.5kg
g	9.8m/s ²
$d_k, (k = x, y, z, \phi, \theta, \psi)$	0.012N·s ² /rad ²
J_x	0.01745kg·m ²
J_y	0.01745kg·m ²
J_z	0.03175kg·m ²
μ	0.8
ub	0.2
lb	-0.2

$A_0, A_1, \bar{A}_0$ and $\bar{A}_1$ are set as

$$A_0 = 16I, A_1 = 8I, \bar{A}_0 = 36I, \bar{A}_1 = 12I.$$

The image of the function based on the absolute value of the distance is drawn in Fig. 4a. And the comparison of convergence curves for testing this function is as in Fig. 4b, where NOA converges faster than the classic particle swarm algorithm (PSO). The trajectory curves are shown in Fig.

5 (NOA) and Fig. 6 (PSO) from which one can conclude that the path quality of NOA is better than PSO. Drones can maintain relatively stable formation and avoid the obstacles safely during the flight to the destination simultaneously. It can be seen that the multi-UAV system maintains relatively small formation errors and continues to approach the target point steadily.

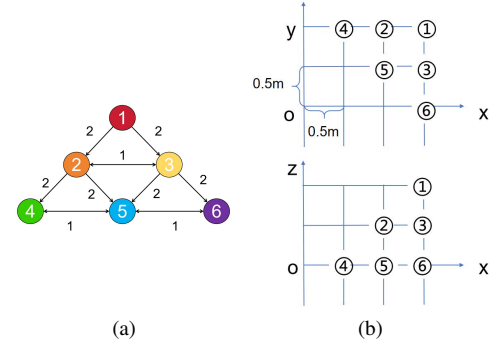


Fig. 3: (a) The communication weight topology diagram between drones; (b) The relative positions of the formation.

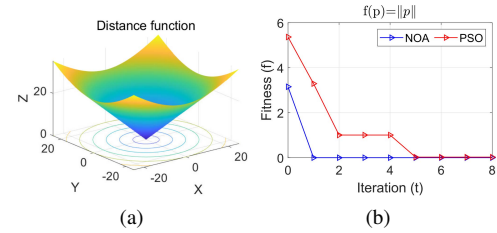


Fig. 4: (a) The distances function; (b) Comparison of convergence curves between NOA and PSO.

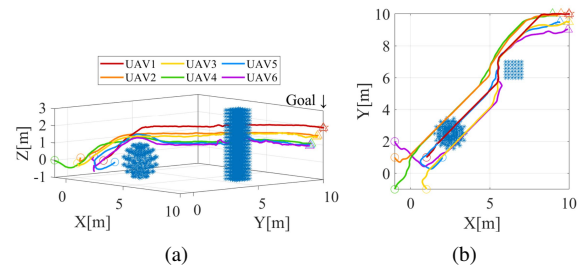


Fig. 5: NOA (a) The trajectories of drones; (b) The top-down view of the trajectories.

5 Conclusion

This paper proposes a novel scheme for path planning and obstacle avoidance strategy through NOA algorithm over the weighted communication topology. Based on the UAV dynamics, the tracking error system is built and driven to zero by the control law designed using FAS technique. Through theoretical analysis and simulation experiments, it has been verified that the proposed strategy and the designed control law can ensure the obstacle avoidance task and the stable convergence of the error formation system. In the future

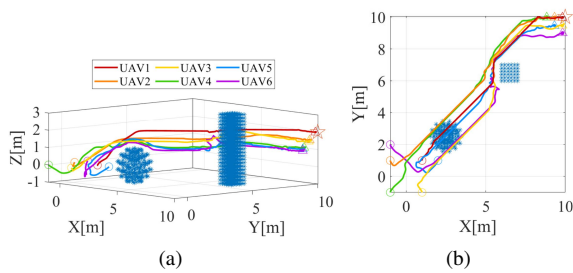


Fig. 6: PSO (a) The trajectories of drones; (b) The top-down view of the trajectories.

work, we will consider the disturbance and fault in multi-UAV system and design observers to estimate and compensate the effects.

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